Benjamin Li

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Education

Cornell University Ithaca, NY
B.A. in Computer Science, Minor in Mathematics; GPA: 3.96 *Expected Dec 2028*
- Relevant Coursework: Object-Oriented Programming and Data Structures, Discrete Structures, Linear Algebra, Multivariable Calculus, Game Theory, Machine Learning, Computer Graphics, Foundations for Robotics

Experience

Cornell Computational Imaging Lab Ithaca, NY
Undergraduate Researcher *Oct 2025 – Present*
- Developing machine learning algorithms for safer microscope imaging of biological samples with reduced light exposure.
- Modeling transformer variants and uncertainty-calibration methods using conformal prediction, bayesian inference, and evidential learning; contributed experiments to tighter conformal calibration with a quantile-conditioned U-Net (accepted NeurIPS 2026 main track) and led research on 5x reduced photon-exposure imaging via uncertainty-aware denoising (to be submitted).

Regeneron Pharmaceuticals Tarrytown, NY
Software Engineering Intern *Jun 2025 – Aug 2025*
- Worked directly with the Regeneron Genetics Center's Chief Data Officer to build and open-source a quantum optimization library for protein-ligand docking, formulated as NP-hard max-weight clique and encoded as QUBO using Python, NumPy, and Qiskit.
- Implemented quantum solvers (QAOA, VQE) and benchmarked them against classical brute-force approaches, finding they produced near-optimal but not consistently optimal solutions; presented results to Regeneron and IBM research teams.

New Jersey Academy of Sciences Remote
Software Engineering Intern *Mar 2025 – May 2025*
- Built full-stack application for judging system that processed results for approximately 200 students and 80 judges at the NJAS research symposium, continued usage and consultation through 2026.
- Shortened formerly hours-long manual result verification to instantaneous feedback through a reusable, open-source workflow featuring a Flask app with a Firebase cloud firestore database used by over a dozen volunteers and event organizers.

Montclair State University Montclair, NJ
Research Intern *Jan 2023 – Nov 2023*
- Developed computer vision framework with PyTorch and NumPy for detecting toxin-producing cyanobacteria, improving on SOTA methods by more than 10% (avg. precision); attended IEEE conference and paper published in proceedings.
- Handled data collection, uniting local data from dept. of environmental science with online iNaturalist data; various methodological improvements, e.g. genetic hyperparameter optimization, specialized data augmentations.

Projects

Evidential Learning for Image Reconstruction
- Implemented novel uncertainty quantification algorithm for grayscale to color image reconstruction via evidential deep learning; wrote related article featured as editor's pick for Towards Data Science, publication with 150,000+ monthly subscribers.

Autograd from Scratch
- Led a four-person team in reverse-engineering PyTorch's automatic differentiation engine used to train ML models. Includes matrix operations, common nonlinearities, SGD, and computation-graph visualization; validated it at 98% MNIST test accuracy.

Learning-Based Cache Eviction
- Worked with 8-person team on an ML eviction policy to improve over deterministic methods; constructed synthetic workloads and model experiments with PyTorch, scikit-learn, and NumPy, improving cache hit rate by up to 8% over the least-recently-used baseline.

Publications

Li, B., Ding, K., Dera, D. (2025). MD-SA2: optimizing Segment Anything 2 for multimodal, depth-aware brain tumor segmentation in sub-Saharan populations. *Journal of Medical Imaging* 12(2). doi:10.1117/1.JMI.12.2.024007

Li, B., Serrano, K., Mazzaro, M., Wu, M., Wang, W., & Zhu, M. (2023). Identification of Cyanobacteria for Harmful Algal Blooms Research Using the YOLO Framework. *IEEE UEMCON*. doi:10.1109/uemcon59035.2023.10316078

Selected Honors

Regeneron Science Talent Search Top 40 Finalist, $25000 award (2025); 1st Place Math and CS, NJ Academy of Sciences Symposium (2024); IEEE UEMCON Best Paper, AI/ML (2023)

Technical Skills

Languages: Python, Java, JavaScript, C/C++, Rust, SQL
Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Qiskit, React.js, Flask, Flutter, Django, Firebase
Tools: Git, Docker, Linux/Unix, Slurm